

Hydrodynamics and Sedimentary Responses Within the Bottom Boundary Layer: Sabine Bank, Western Louisiana

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Abstract

To determine the wave-current interactions and bottom boundary layer characteristics of eastern Sabine Bank, off the southwestern Louisiana coast, an extensive array of bottom mounted equipment was deployed for a period of 44 days (11 March - 23 April, 2004). The instrument array mainly consisted of Acoustic Doppler Velocimeters (ADV's), Electromagnetic current meters (ECM's), Optical Backscatters (OBS's) and pressure sensors deployed at three locations (on the crest along the eastern flank of the bank, offshore and onshore of the bank). The results show that waves were generally from south-east with a mean peak period of 5.2 s. The highest significant wave height obtained was 1.8 m. Velocities parallel to the coast prevailed during fair-weather wave conditions. However, during cold fronts and strong wind regimes, cross-shore velocity components dominated. The significant wave height and the corresponding shear stress and shear velocity, all show higher values associated with the passage of cold fronts and during strong southerly/southeasterly wind regimes, when larger than the threshold for sediment re-suspension velocities were estimated from grain-size data. Hence it is inferred that waves during fair weather are too weak to re-suspend bottom sediments. However, waves during the cold fronts can effectively re-suspend and re-distribute sediments along the bank. It is also suggested that the strong wind regimes which usually occur during this period of the year can strongly affect sediment transport as effectively as the cold fronts do.

Introduction

The Louisiana coast has several unique characteristics in terms of coastal processes. First, it is a low-energy environment and hence waves and tidal currents are usually weak in the region. Secondly, the Mississippi River, which is one of the largest river systems in the world, discharges a tremendous amount of sediment onto the Gulf of Mexico continental shelf. Third, the coast is exposed to frequent cold fronts between October and April, and occasional storms and hurricanes during summer (Stone et al., 1997; Muller and Stone, 2001). Additionally, the relative sea-level rise is occurring at one of the highest rates in the northern hemisphere (Penland and Ramsey, 1990).

Cold fronts and tropical storms have significant impacts on coastal processes. Studies of wave-climate and bottom boundary layer interactions have been undertaken along the Louisiana coast since the 1980's (Adams et al., 1987; Friedrichs et al., 2000; Grant and Madsen, 1986; Pepper and Stone, 2002; 2004; Stone, 2000; Wright et al., 1997). These studies show that in the northern Gulf of Mexico, waves are too small to re-suspend sediments during fair weather. Wave induced sediment re-suspension during cold fronts, storms and hurricanes causes significant impacts on the bottom boundary layer and subsequently sediment transport. However, studies relating the wave-climate and sediment transport within the bottom boundary layer have not been undertaken along the western Louisiana coast except for a few qualitative studies (Roberts et al., 1989; Crout and Hamiter, 1981).

This paper is focused on a brief introduction of the wave-current-bottom boundary layer processes based on field data collected from March through April, 2004. The study site is offshore western Louisiana at the 10 m isobath.

Background

Sabine Bank is a submerged bank located 28 km offshore from Sabine Pass and spans 50 km in maximum length and 7 km in maximum width (Fig. 1). There are also several other submerged banks in its vicinity which were formed due to rapid sea level rise during the late Holocene, as remnants of beach ridges (Rodriguez et al., 1999; 2004). In western Louisiana, a wide range of research pertaining to the evolution of the submerged banks has been undertaken based on extensive core data (Anderson and Wellner, 2002; Rodriguez et al., 1999; 2004). Research pertaining to wave climate and bottom boundary layer dynamics has not been undertaken to date.

Methodology

Field work was undertaken to measure waves, currents and bottom boundary layer interactions on eastern Sabine Bank from 11 March - 23 April, 2004. The primary instrument array consisted of two Acoustic Doppler Velocimeters (ADV's) and a unique instrumentation array named WADMAS (Stone, 2000) which consisted of three Electromagnetic current meters (ECM's), three Optical Backscatters (OBS's) and pressure sensors deployed at three locations: WADMAS on the crest of eastern Sabine Bank and two ADV's onshore and offshore of the bank, respectively (Fig. 1).

Meteorological data were obtained from a nearby National Data Buoy Center (NDBC) buoy station, SRST2 (29°40.2'N, 94°03'W). Data on surface currents from the nearby eastern Sabine Bank were obtained from TABS (Texas Automated Buoy System) Station R (29°38.1'N, 93°38.502'W) (Guinasso et al., 2001) (Fig.1).

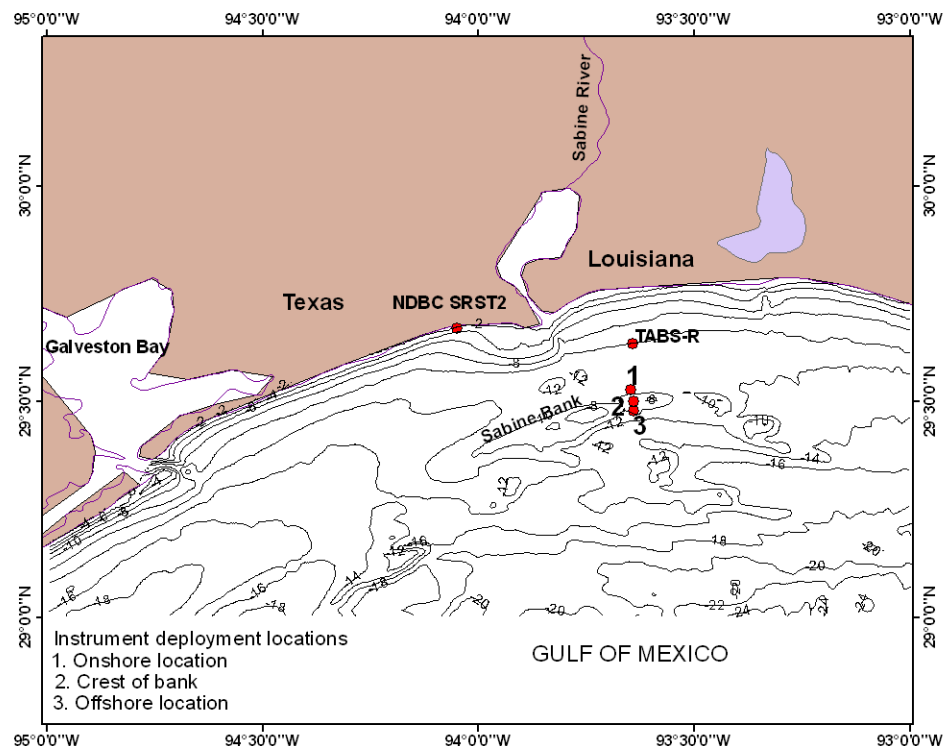


Figure 1. Map of Sabine Bank, offshore south-west Louisiana.

Seasonal Weather Patterns Along Northern Gulf of Mexico

The northern Gulf of Mexico experiences frequent cold front events between October and April and occasional storms and hurricane passages especially in the summer; all of these systems have significant impacts on coastal processes (Roberts et al., 1989; Stone, 2000). In south-west Louisiana, a total of 14 tropical storms and hurricanes have impacted the coast from 1901 to 1996 (Stone et al., 1997). However, although they are weaker than tropical storms and hurricanes, cold fronts may be more important because they are generated more frequently (i.e., 20 to 30 times a year).

Hsu (1988) and Roberts et al. (1989) provide detailed explanations for the mechanism of cold front generation, sustenance and dissipation in the northern Gulf of Mexico. Along the Louisiana coast, winds usually blow from south/south-east; however, once cold fronts pass the area, wind direction dramatically changes from the southern quadrants to from the north, wind speed increases, and air temperature drops (Fig. 2). Roberts et al. (1989) shows that two types of cold fronts exist in Louisiana: parallel case and oblique case, subject to the initial location of air masses and how the weather systems grow.

Relatively strong southerly/southeasterly winds are characterized between the periods of cold fronts (Fig. 3). The weather charts from National Center for Environmental Prediction (NCEP) and meteorological data from NDBC during the survey period verified these trends (Fig. 3).

Waves and Currents Along the Western Louisiana Coast

Figure 2 shows time series plots of wave parameters, currents, and OBS's during the entire study period. Waves usually propagated from southern quadrants; however, during cold fronts, waves from the north are apparent. Waves are strongly influenced by the meteorological conditions such as wind speed and wind direction. Maximum significant wave height was 1.8 m, from the south.

The coastal current data from the onshore station (station 2) and TABS station R (top figure) are shown in Fig. 2 (Guinasso et al., 2001). It is interesting to note that the coastal currents during the observation period are predominantly to the west contrary to the north-easterly current prevalent near the bottom, for the onshore study site (Fig. 2). Specifically, the cold fronts and strong southerly wind regime influence the coastal currents in the same manner as it influences the Sabine Bank circulation. The possible generation of shallow water eddies in the region due to the interaction of coastal currents with intermittent shoals and shallow banks near the coast are poorly understood.

Hydrodynamics and Sedimentary Responses Within the Bottom Boundary Layer

The survey results show that waves and currents are strongly associated with weather conditions and effectively influence the bottom boundary layer. The results represent conspicuous characteristics of bottom boundary layer dynamics based on the three meteorological conditions: (1) fair weather, (2) cold fronts, and (3) strong southerly/southeasterly wind regimes. Below wave-current-bottom boundary layer interactions are summarized based on these meteorological conditions.

Fair-weather condition

Figure 3 shows time series plots of bottom boundary layer parameters during fair weather conditions. In the survey area significant wave height was less than 1 m during this period. Maximum shear stress due to waves during this condition is less than 0.16 N/m^2 , which is the computed threshold for the re-suspension of the bottom sediments. Also, shear stress due to currents is mostly less than the threshold value (Fig. 3). It is clear that waves and currents are not strong enough to re-suspend sediment during fair weather conditions.

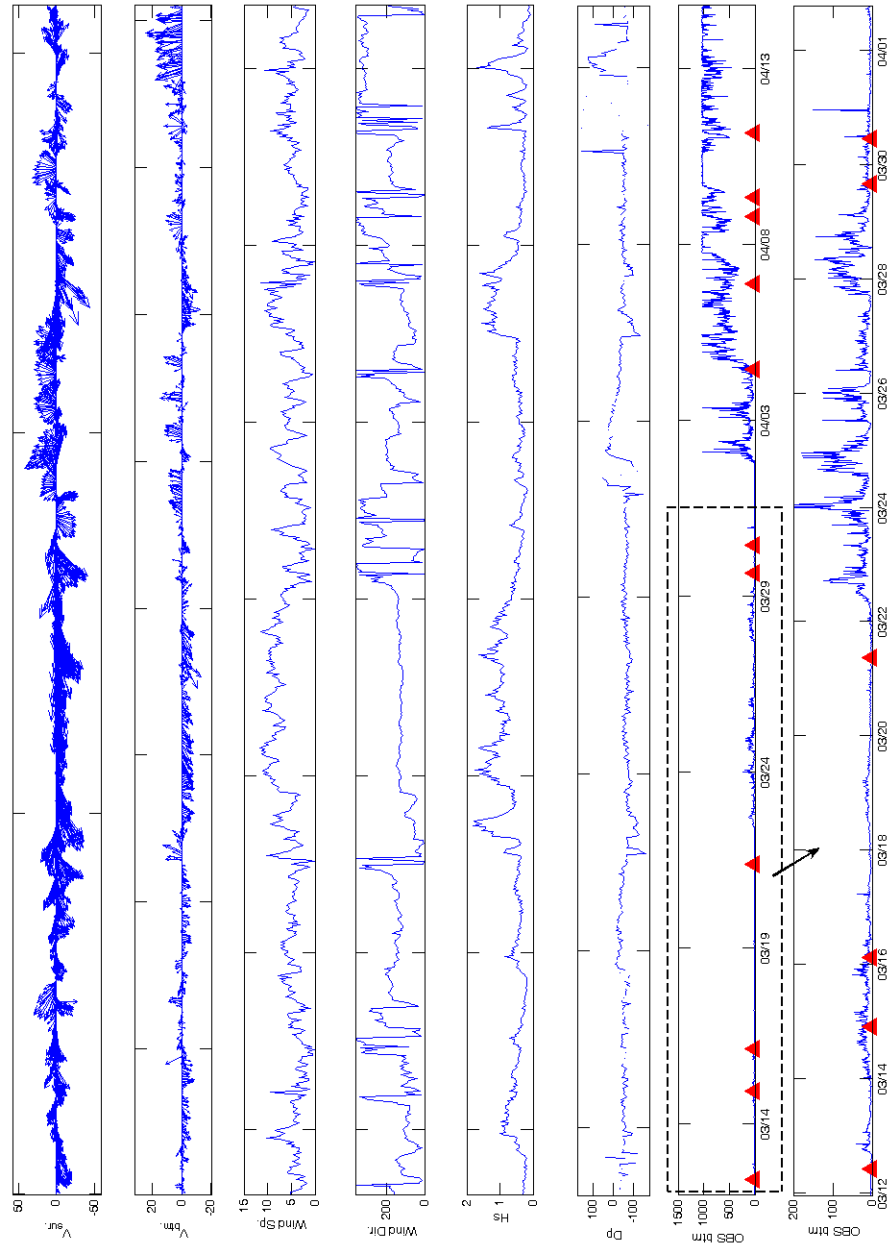


Figure 2. Time series plots of waves, surface and bottom currents, and bottom OBS data. Where V_{sur} is surface current(m/s), V_{btm} is bottom current(m/s), H_s is significant wave height(m), DWD is Dominant wave direction(degree), $Wind\ sp$ is wind speed(m/s), $Wind\ Dir.$ is wind direction(degree) and OBS_{btm} is bottom OBS(NTU). The bottom figure is the OBS result subtracted from the period covered with the dashed rectangle. Red shaded triangles indicate the passage of cold fronts. Surface current was obtained from TABS station R (Guinasso et al., 2001).

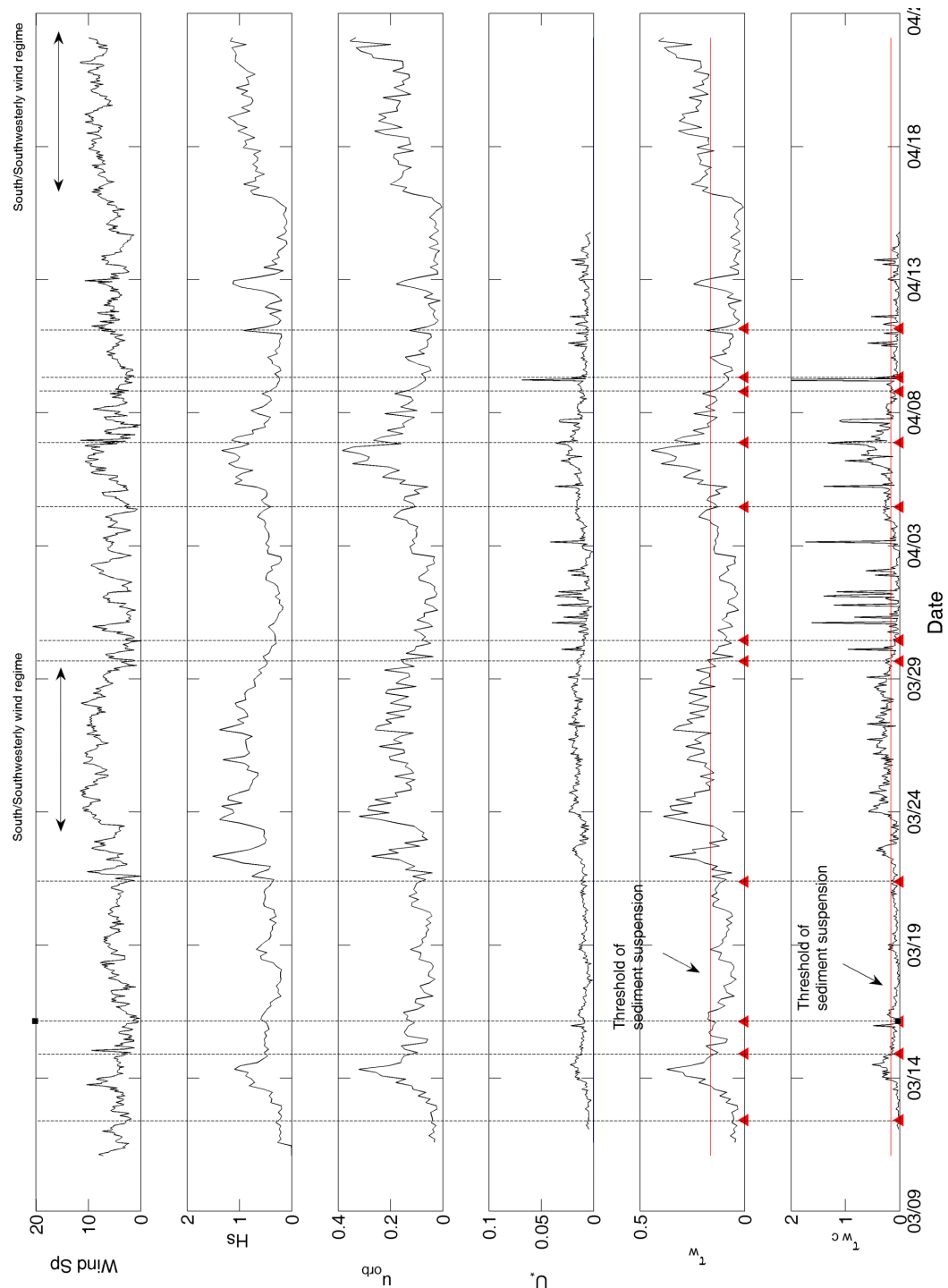


Figure 3. Time series plots of meteorological data obtained from a nearby NDBC buoy station, significant wave height and bottom boundary layer parameters. Where Wind Sp. is wind speed(m/s), Hs is significant wave height(m), uorb is orbital velocity(m/s), U^* is shear velocity(m/s), w is shear stress due to waves, w_c is shear stress due to waves and currents. Dashed lines and red shaded triangles indicate the passage of cold fronts and the threshold of sediment suspension is shown in the figure.

Cold fronts

In Louisiana, waves and bottom boundary layer dynamics are strongly affected by cold fronts as well as storms (Stone, 2000; Stone et al., 2004). During our survey period, at least 10 cold fronts hit the area within a period of 44 days. When we compared the meteorological data with wave and bottom boundary parameters most of the significant wave height (Hs), shear velocity, shear stress and OBS were associated with cold front passages. During the passage of cold fronts, the shear stresses due to both waves and currents were greater than the threshold of the sediment re-suspension (Fig. 3). Hence, it is suggested that waves and currents can effectively re-suspend the sediment during the cold fronts.

Strong southerly/southeasterly wind regimes

Some of the wave and bottom boundary layer parameters did not correspond to the cold front passages but did relate to the strong southerly/southeasterly wind regimes (i.e. from Mar. 22nd to 30th and from Apr. 15th to Apr. 23rd). Figures 2 and 3 show time series plots of wind speed, Hs, shear stress, and OBS data from March 22nd to March 29th corresponding to a strong wind event in the study area. Significant wave height, shear velocity, shear stress, and OBS had maximum values during this period, for the entire survey duration. According to the weather maps from NCEP, during the above period the strong winds from the south or southeast persisted for several days. These wind regimes were associated with the high shear velocity, shear stress and OBS readings. In particular, OBS data showed a maximum value recorded on March 24th, during a period of strong southerly wind (Fig. 2).

These results suggest that during fair weather, waves are too weak to re-suspend sediments. During the high wind speeds elevated significant wave height was caused by persistent strong southerly winds. The associated high shear velocity and shear stress, which were higher than the threshold for sediment re-suspension, correspond to bottom boundary layer conditions during the cold front events. Hence, both waves and currents can easily re-suspend sediment during this period as well as during the cold front events.

It is likely that this situation is not necessarily confined to the western Louisiana coast. The wind, waves and OBS data obtained from a WAVCIS (Wave-Current-Surge Information System) station, CSI-3 during our survey period show similar results during March 22nd to March 30th (Stone et al., 2001). It is inferred that this situation is a common seasonal characteristic all along coastal Louisiana. Crout and Hamiter (1981) showed the relationship between seasonal wind variations and sediment transport, based on the velocity data and a criterion developed by Komar and Miller (1975) for the western Louisiana inner shelf. Although our survey results show that this greatly affects bottom boundary layers, no significant mechanisms have been discussed to date. We are also not attempting any detailed discussion at this point due to the shortage of data from our pilot study. Additional data need to be collected, especially on sediment and bed form characteristics (ripple height and wavelength) as well as supplemental wave and current data so that we can correlate the high wind events with the sediment transport processes quantitatively.

Conclusion

To determine wave characteristics and bottom boundary layer dynamics, a field survey was undertaken on eastern Sabine bank, offshore of south-west Louisiana from March 11 to April 23, 2004 and the following results were obtained.

1. Waves were weak and did not re-suspend sediment during fair weather.
2. Currents were sufficiently strong to re-suspend sediment during the entire period except fair weather.
3. Cold fronts were strongly associated with wave and bottom boundary layer interactions.
4. Strong southerly/southeasterly wind regimes also affected the wave and the bottom boundary layer interactions during the observation period.

In summary, it is suggested that while waves are not an important factor for sediment transport during fair weather conditions, those during cold fronts and strong southerly/southeasterly wind regimes

affect sediment transport significantly. Currents are effectively strong enough to re-suspend sediment for the entire period except during fair-weather. Therefore, it is strongly suggested that the cold fronts and also the strong wind regimes are important to the bottom boundary layer dynamics of the Sabine Bank.

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